Computer fundamental

PRATHAM

UNIT: 1 INTRODUCTION TO COMPUTER

1 . WHAT IS COMPUTER?

- **The word "computer" is comes from the word "TO COMPUTE" means to calculate.**
- **A computer is normally considered to be a calculation device which can perform the arithmetic operations very speedily.**
- **A computer may be defined as a device which operates upon the data.**
- **Data can be in the form of numbers, letters, symbols, size etc. And it comes in various shapes & sizes depending upon the type of computer application.**
- **A computer can store, process & retrieve data as and when we desired.**
- **The fact that computer process data is so fundamental that many people have started calling as "Data Processor".**
- **A computer first it gets the Data, does Process on it and then produces Information.**

DATA PROCESS INFORMATION

- **DEFINATION OF COMPUTER**
  - **A computer is an electronic device which takes input from the user, processes it and gives the output as per user's requirement.**
  - **So the main tasks of performed by the computer are:**
    - **Input**
    - **Process**
    - **Output**

2 . WRITE DOWN THE CHARACTERISTICS OF COMPUTER

Some important characteristics of the computer are as follow:

- **Automatic:**
  - **Computers are automatic machines because it works by itself without human intervention.**
  - **Once it started on a job they carry on until the job is finished.**
  - **Computer cannot start themselves.**

- **They can works from the instructions which are stored inside the system in the form of programs which specify how a particular job is to be done.**

- **Accuracy:**
    - **The accuracy of a computer is very high.**
    - **The degree of accuracy of a particular computer depends upon its design.**
    - **Errors can occur by the computer. But these are due to human weakness, due to incorrect data, but not due to the technological weakness.**
- **Speed:**
    - **Computer is a very fact device. It can perform the amount of work in few seconds for which a human can take an entire year.**
    - **While talking about computer speed we do not talk in terms of seconds and milliseconds but in microseconds.**
    - **A powerful computer is capable of performing several billion (109) simple arithmetic operations per second.**
- **Diligence:**
    - **Unlike human beings, a computer is free from monotony, tiredness & lack of concentration.**
    - **It can continuously work for hours without creating any error & without grumbling.**
    - **If you give ten million calculations to performed, it will perform with exactly the same accuracy & speed as the first one.**
- **Versatility:**
    - **It is one of the most wonderful features about the computer.**
    - **One moment it is preparing the results of a particular examination, the next moment it is busy with preparing electricity bills and in between it may be helping an office secretary to trace an important letter in seconds.**
- **Power of remembering:**
    - **Computer can store and recall any amount of data because of its high storage capacity of its storage devices.**
    - **Every piece of information can be retained as long as desired by the user and can be recalled as and when required.**
    - **Even after several years, if the information recalled, it will be as accurate as on the day when it was filled to the computers.**
- **No I.Q.**
    - **A computer is not a magical device; it processes no intelligence of its own.**
    - **Its I.Q. is zero.**

- It has to be told what to do & in what sequence.
- It cannot take its own decision.

- **No Fallings:**
    - A Computer has no feelings because they are machines.
    - Based on our feelings, task, knowledge and experience we often make certain judgments in our day today life.
    - But Computer goes exactly the way which we have given the instructions.

3. EXPLAIN THE DATA PROCESSING CYCLE OF COMPUTER.

- The computer Data Processing is any process that a computer program does to enter data & summaries, analyse or convert data into useable information.
- The process may be automated & run on a computer.
- It involves recording, analyzing, storing, summarizing & storing data.
- Because data are most useful when it is well presented & informative.

<u>The Data Processing Cycle:</u>

- Data Processing cycle described all activities which are common to all data processing systems from manual to electronic systems.
- These activities can be grouped in four functional categories, viz., data input, data processing, data output and storage, constituting what is known as a data processing cycle.
- The main aim of data processing cycle is to convert the data into meaningful information.
- Data processing system are often referred to as Information System.
- The Information System typically take raw Data as Input to produce Information as Output.

- The data processing cycle contains main four functions:
    - Data input

    - Data process
    - Data storage
    - Data output

- <u>DATA INPUT</u>
    - The term input refers to the activities required to record data.
    - It's a process to entered data in to computer system.
    - So before we input any data, it is necessary to check or verify the data context.
- <u>DATA PROCESSING</u>
    - The term processing includes the activities like classifying, storing, calculating, comparing or summarising the data.
    - The processing means to use techniques to convert the data into meaningful information.
- <u>DATA OUTPUT</u>
    - It's a communication function which transmits the information to the outside world.
    - After completed the process the data are converted into the meaningful in
    - Sometimes the output also includes the decoding activity which converts the electronically generated information into human readable form.
- <u>DATA STORAGE</u>
    - It involves the filling of data & information for future use.

4. EXPLAIN TH E CLASSIFICATION OF THE COMPUTER BY DATA PROCESSED

The computers are divided mainly three types on the based on data processed:

1. Analog computers
2. Digital computers
3. Hybrid computers

<u>Analog computers:</u>

- In Analog Computers, data is represented as continuously varying voltage and operate essentially by measuring rather counting.
- As the data is continuously variable, the results obtained are estimated and not exactly repeatable.
- It can able to perform multiple tasks simultaneously and also capable to work effectively with the irrational number. E.g. 1/8 = 0.125 and 1/6=0.1666

- Voltage, temperature and pressure are measured using analog devices like voltmeters, thermometers and barometers.

Digital Computers

- The digit computer is a machine based on digital technology which represents information by numerical digit.
- In Digital Computers data is represented as discrete units of electrical pulses. The data is measured in quantities represented as either the 'on' or 'off' state.
- Therefore, the results obtained from a digital computer are accurate.
- Virtually all of today's computers are based on digital computers.

Hybrid Computers

- It combines the good features of both analog & digital computers.
- It has a speed of analog computer & accuracy of digital computer.
- Hybrid Computers accept data in analog form and present output also in digitally.
- The data however is processed digitally.
- Therefore, hybrid computers require analog-to-digital and digital-to-analog converters for output.

5. EXPLAIN THE CLASSIFICATION OF THE COMPUTER BY DATA PROCESSING:

The computers are classified in four types on the based on data processing.

- Micro computer
- Mini computer
- Mainframe computer
- Super computer

Micro Computer:

- Micro computers are the computers with having a microprocessor chip as it central processing unit.
- Originated in late 1970s.
- First micro computer was built with 8 bit processor.
- Microcomputer is known as personal computer.
- Designed to use by individual whether in the form of pc's, workstation or notebook computers.
- Small in size and affordable for general people.
- Ex: IBM PC, IBM PC/XT, IBM PC/AT

Micro Computer:

- Mini computers are originated in 1960s.
- Small mainframes that perform limited tasks.
- Less expensive than mainframe computer.
- Mini computers are Lower mainframe in the terms of processing capabilities.
- Capable of supporting 10 to 100 users simultaneously.
- In 1970s it contains 8 bit or 12 bit processor.
- Gradually the architecture requirement is grown and 16 and 32 bit.
- Minicomputers are invented which are known as supermini computers.
- Ex: IBM AS400

Mainframe Computer:

- A very powerful computer which capable of supporting thousands of user simultaneously.
- It contains powerful data processing system.
- It is capable to run multiple operating systems.
- It is capable to process 100 million instructions per second.
- Mainframes are very large & expensive computers with having larger internal storage capacity & high processing speed.
- Mainframes are used in the organization that need to process large number of transaction online & required a computer system having massive storage & processing capabilities.
- Mainly used to handle bulk of data & information for processing.
- Mainframe system is housed in a central location with several user terminal connected to it.
- Much bigger in size & needs a large rooms with closely humidity & temperature.
- IBM & DEC are major vendors of mainframes.
- Ex : MEDHA, SPERRY, IBM, DEC, HP, HCL

Super Computer:

- Most powerful & most expensive computer.
- Used for complex scientific application that requires huge processing power.
- Used multiprocessor technology to perform the calculation very speedy.
- They are special purpose computers that are designed to perform some specific task.

- The cost of the super computer is depended on its processing capabilities & configuration.
- The speed of modern computer is measured in gigaflops, teraflops and petaflops.
  - Gigaflops= 109 arithmetic operation per second.
  - Teraflops=1012 arithmetic operation per second.
  - Petaflops=1015 arithmetic operation per second.
- Ex: PARAM , EKA, BLUE GENE/P

6. EXPLAIN THE GENERATION OF THE COMPUTERS.

In Computer language, "Generation" is a set of Technology. It provides a framework for the growth of the computer technology. There are totally Five Computer Generations till today. Discussed as following.

First Generation:

- Duration: 1942-1955
- Technology: vacuum tube
  - Used as a calculating device.
  - Performed calculations in milliseconds.
  - To bulky in size & complex design.
  - Required large room to place it.
  - Generates too much heat & burnt.
  - Required continuously hardware maintenance.
  - Generates much heat so must air-conditioner rooms are required.
  - Commercial production is difficult & costly.
  - Difficult to configure.
  - Limited commercial use.
  - ENIAC, EDVAC, EDSAC are example of 1st generation computer.

Second Generation:

- Duration: 1955-1964
- Technology: transistor
  - 10 times Smaller in size than 1st generation system.
  - Less heat than 1st generation computers.
  - Consumed less power than 1st generation system.
  - Computers were done calculations in microseconds.
  - Air-conditioner is also required.
  - Easy to configure than 1st generation computers.

- More reliable in information.
- Wider commercial use.
- Large & fast primary/secondary storage than 1st generation computers.

Third Generation:

- Duration: 1965-1975
- Technology: IC chip
  - Smaller in size than 1st & 2nd generation computers.
  - Perform more fast calculations than 2nd generation systems.
  - Large & fast primary/secondary storage than 2nd generation computers.
  - Air –conditioner is required.
  - Widely used for commercial applications.
  - General purpose computers.
  - High level languages like COBOL & FORTAN are allowed to write programs.
  - Generate less heat & consumed less power than 2nd generation computer.

Fourth Generation:

- Duration: 1975-1989
- Technology: Microprocessor chip
  - Based on LSI & VLSI microprocessor chip.
  - Smaller in size.
  - Much faster than previous generations.
  - Minimum hardware maintenance is required.
  - Very reliable as computer to previous generation computers.
  - Totally general purpose computer.
  - Easy to configure.
  - Possible to use network concept to connect the computer together.
  - NO requirement of air-conditioners.
  - Cheapest in price.

Fifth Generation:

- Duration: 1989 to Present
- Technology: ULSI microprocessor chip
  - Much smaller & handy.
  - Based on the ULSI chip which contains 100 million electronic components.
  - The speed of the operations is increased.
  - Consumed less power.
  - Air-conditioner is not required.
  - More user friendly interface with multi-media features.
  - High level languages are allowed to write programs.

- Larger & faster primary/secondary storage than previous generations.

- Notebook computers are the example of 5th generation computers.

7. . EXPLAIN THE BLOCK DIAGRAM OF COMPUTER OR EXPLAIN THE SIMPLE MODEL COMPUTER.

A simple computer system comprises the basic components like Input Devices, CPU (Central Processing Unit) and Output Devices as under:

- **Input Devices:**
  - The devices which are used to entered data in the computer systems are known as input devices.
  - Keyboard, mouse, scanner, mike, light pen etc are example of input devices.

FUNCTION OF INPUT DEVICES

  - Accept the data from the outside worlds.
  - Convert that data into computer coded information.
  - Supply this data to CPU for further processing.
- **Output Devices:**
  - The devices which display the result generated by the computer are known as output devices.
  - Monitor, printer, plotter, speaker etc are the example of output devices.

FUNCTIONS OF OUTPUT DEVICES

  - Accept the result form the CPU.
  - Convert that result into human readable form.

- Display the result on the output device.

- **Memory Unit:**
  - The data & instruction have to store inside the computer before the actual processing start.
  - Same way the result of the computer must be stored before passed to the output devices. This tasks performed by memory unit.

FUNCTIONS OF MEMORY UNIT

- Store data & instruction received from input devices.
- Store the intermediate results generated by CPU.
- Store the final result generated by CPU.

- **Arithmetical & Logical Unit:**
  - The ALU is the place where actual data & instruction are processed.
  - All the calculations are performed & all comparisons are made in ALU.
  - Performs all arithmetical & logical operations.
  - An arithmetic operation contains basic operations like addition, subtraction, multiplication, division.
  - Logical operations contains comparison such as less than, greater than, less than equal to, greater than equal to, equal to, not equal to.

- **Control Unit:**
  - It controls the movement of data and program instructions into and out of the CPU, and to control the operations of the ALU.
  - In sort, its main function is to manage all the activities within the computer system.
  - Controls the internal parts as well as the external parts related with the computer.
- **CPU:**
  - The Unit where all the processing is done is called as Central Processing Unit.
  - It contains many other units under it.
  - Main of them are:- Control Unit And ALU (Arithmetic & Logic Unit)

UNIT: 2 INPUT DEVICES

1 . WHAT IS INPUT DEVICES ?

- **The Input devices are the devices which are used to enter the data in the computer system.**
- **Keyboard, mouse, scanner, microphone are the example of input devices.**

FUNCTIONS OF INPUT DEVICES:

- **Accept the data from the outside worlds.**
- **Convert that data into computer coded information.**
- **Supply this data to Central Processing Unit for further processing.**

CLASIFICATION OF INPUT DEVICES:

INPUT DEVICES

STANDARD INPUT DEVICE	POINTING DEVICES	SPECIAL INPUT DEVICES
KEYBOARD	MOUSE	MICROPHONE
	TRACK BALL	SCANNER
	JOYSTICK	CAMERA
	LIGHT PEN	
	TOUCH SCREEN	

2 . EXPLAIN STANDARD INPUT DEVICE: KEYBOARD.

- **Keyboard is most commonly used input device.**
- **It is similar like a type writer which is used to enter data in the computer.**
- **It contains sets of keys such as alphabets, number & special signs.**
- **There are two types of keyboard.**
    - **General purpose keyboard**
    - **Special purpose keyboard**

GENERAL PURPOSE KEYBOARD:

- **Standard keyboard which are used in personal computers.**
- **It contains enough keys which are used in all types of applications so they are known as general purpose keyboard.**
- **Most popular general purpose keyboard contains 101 keys.**
- **The general purpose keyboard are divided into following parts:**

ALPHANUMERAIC KEYPAD

- **The centred part of the keyboard is known as alphanumeric keypad.**
- **It contains alphabets, numbers & special signs such as *,!, @, #, $, %,* etc.**

NUMERAIC KEYPAD

- **The right most part of the keyboard is known as numeric keypad.**
- **It contains 0 to 9 numbers & mathematical signs such as +, *, -, /.**
- **Mainly used for fast data entry in mathematical applications.**

ARROW KEYS

- **Set of four keys up, down, left & right.**
- **Used to move the cursor at left & right or up and down on the screen.**
- **They are referred as “cursor-control” or “cursor-movement” keys.**

FUNCTION KEYS

- **The first line of the keyboard contains a Set of 12 keys with name f1 to f2 are known as function keys.**
- **Used to generate short-cuts in different software package.**

SPECIAL KEYS

- **There are lots of keys that are used for some specific task describes follows:**
- **TAB: used for gives multiple spaces or move the cursor to next defined position.**
- **ENTER: used for generate the output of any command.**

- SPACE: used to make one blank space between two words.
- BACKSPACE: used to remove the left-most character at cursor position.
- DELETE: used to remove the right-most character at cursor position.
- HOME: moves the cursor at the beginning of the line.
- END: moves cursor at the end of the line.
- PAGE UP: moves or scroll the screen up or previous page of the current page.
- PAGE DOWN: moves the screen to the next page from the currently displayed page.
- PRINT SCREEN: used to print what is currently displayed on the screen.
- INSERT: used to enter text between two characters.
- ESC: used to negate current command or terminate the execution of the program.
- ALT: used to expand the functionality of keyboard. Basically used to generate shortcuts in different application.
- CTRL: used to expand the functionality of keyboard. Basically used to generate shortcuts in different application.
- NUMLOCK: used to on or off the numeric keypad.
- CAPSLOCK: used to type the all inputted text capitally.

SPECIAL PURPOSE KEYBOARD

- Special purpose keyboard is used for special purpose applications which required faster data entry and rapid interaction with the computer system.
- For example ATM used in banks used special purpose keyboard which contains a few keys.

3 . EXPLAIN POINTING DEVICES.

1 . MOUSE

- Mouse is Small hand-hold device Input device which is generally used for drawing purpose.
- It's a Pointing device.
- It contains two or three buttons
- Left button is used to point out or select any item by clicking.
- Right to generate context menu.
- When user moves mouse across flat surface, the graphic cursor moves on screen.
- Graphic cursor contains verity of symbols such as arrow, wrist, pointing finger etc.

- Depending on application text & graphic cursors are changed.
- The following 5 techniques are used to carry out various operations:
- POINT:
    - To move the mouse on top of icon
- C LICK:
    - To press & release the left button of mouse at once.
    - Used to open any currently selected icon, menu.
- DOUBLE CLICK:
    - To press & release the left button of mouse twice.
    - Used to open any application or program.

SIMULTANEOUS-CLICK:

- Press & release left & right button to gather.
- Used in some software package to added some functionality.

- DRAG:
    - Press the left button down & moved the mouse on screen.
    - Used to move the graphics on screen.
- Many types of mouse are available such as mechanical mouse, optical mouse, serial mouse, wireless mouse which are used for different purpose.

2 . TRACK BALL

- Trackball is a pointing device which is similar to a mouse.
- A ball is placed on the track ball device which is used to move the graphic cursor on the screen.
- It also contains buttons which are used to select a particular item on the screen.
- To move the graphic cursor on screen, the ball is rolled with the fingers or thumb.

- It needs not to move the whole device to move the cursor so it is often attached with some keyboards.
- Track balls come in various shapes with same functionality.
- Commonly three shapes are used: ABALL, A SQUARE, and A SQUARE.
- In case of ball we need to move it with the help of finger.
- In case of button pushed with finger in desired direction of the cursor movement.
- In case of button press finger to up or down & left or right to move cursor.

<u>Advantages of track ball</u>

  - Takes less desk space.

- Takes less arm movements than mouse.
- Doesn't require any mouse pad & large area to move the mouse.
- Less strain on the wrist.
- Finger trip control which may offer more accuracy than mouse.

3. JOYSTICK

- Joystick is a pointing device which is works on the same principle of track ball.
- It contains a stick which is placed on the spherical ball.
- The stick is used to move the cursor at desired position left or right or backwardor forward.
- It also contain button that is clicked to make selection of currently pointed item.
- A joystick is similar to a mouse, except that with a mouse the cursor stopsmoving as soon as you stop moving the mouse.
- With a joystick, the pointer continues moving in the direction the joystick ispointing.
- To stop the pointer, you must return the joystick to its upright position.
- Some of the systems using joysticks are
    - Aircrafts, UAVs for flight control
    - Motorized Wheelchairs as input device
    - Microscopes
    - Submarines
    - Security Systems
    - Video Games
- Joysticks are widely used for video games
- <u>Advantages of joystick</u>
- It is very easy to learn to use.